Identifying Obstacles to Equitable Flood Mitigation Funding

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Abstract

As damages from floods, fires, and other climate extremes mount, public investments in risk reduction have also grown. In the US, public spending on risk reduction efforts totaled over $X in 2024, and demand for such funds regularly exceeds available funding. Evidence suggests that in this competitive landscape for funding, urban and whiter communities have fared better at obtaining funding but are often spending it on more disadvantaged neighborhoods within them. However, the reasons for this trend are unclear: complexities in the application process, cost-benefit analysis requirements, and flood damage may all play a role in driving the uneven funding patterns. Here, using a novel dataset tracking properties from flood exposure to application to funding receipt, we identify how each hurdle in the mitigation process shapes the properties that ultimately benefit. Our results demonstrate that the application stage is the critical pinch point in the mitigation funding pipeline: of all flooded properties, 67.5% are eligible for public assistance and 61% are located in communities where local governments have submitted applications, yet only 2.3% apply and approximately 1% are ultimately funded. Homeowners in less-white neighborhoods and more affordable homes relative to their neighbors are more likely to apply, and this trend deepens for those who ultimately receive funding, while higher flood exposure has minimal correlation — implying that funding may not reach all who need it or may be interested.

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1. Introduction

Flooding is the most costly natural hazard in the United States, and projected climate changes will intensify both its frequency and its financial burden (for Environmental Information, 2024). An estimated 41 million people live within the 100-year floodplain, more than twice the figure suggested by FEMA's official flood maps, and the populations most exposed are disproportionately low-income, Black, and Native American (Tate et al., 2021). In response to mounting losses, federal investment in flood risk reduction has grown substantially, averaging over $1.2 billion per year in FEMA hazard mitigation grants alone over the last decade. Yet growing evidence suggests this investment is not reaching those who need it most. Nationwide analyses find that wealthier and whiter counties have secured more federal mitigation funding, while racially and economically marginalized communities, which bear a disproportionate flood burden, have received comparatively less (Mach et al., 2019; Elliott et al., 2020; Loughran and Elliott, 2022). Several mechanisms have been proposed to explain this pattern: federal cost-benefit analysis requirements that systematically favor high-value properties, limited local government capacity to administer grants, and lack of household awareness about available funding. Prior research, however, has been constrained by a fundamental data limitation: we can observe only who *receives* funding, not who is eligible or who applies. As a result, it remains unclear where in the process inequities first emerge and which interventions would most effectively improve access.

Here, we combine novel datasets spanning over two decades of flood exposure, grant applications, and funding allocations in North Carolina to examine the full flooded-to-funded pipeline (Fig. 1). Flood exposure is derived from a random forest model trained on address-level National Flood Insurance Program (NFIP) records, producing inundation footprints for 78 damaging events between 1996 and 2020 (Garcia et al., 2025). We link these exposure records to property-level application and funding data for FEMA Hazard Mitigation Assistance (HMA), provided by the North Carolina Department of Public Safety. Together, these datasets allow us to track individual properties across five stages (flooded, eligible, in an applying community, applied, and funded) and to examine how the composition of properties shifts in terms of flood exposure history, property value, and neighborhood racial demographics at each transition. This is the first analysis to follow properties from flood exposure through to mitigation funding outcome at the property level, making it possible to distinguish between households that are not being selected and those that never enter the applicant pool at all.

1.1. Access to federal funding for climate resilience

The federal government is the primary backstop against flood losses in the United States, operating programs spanning pre-event risk reduction, immediate post-disaster relief, and longer-term recovery. We focus our analysis on the Federal Emergency Management Agency (FEMA), which leads disaster response

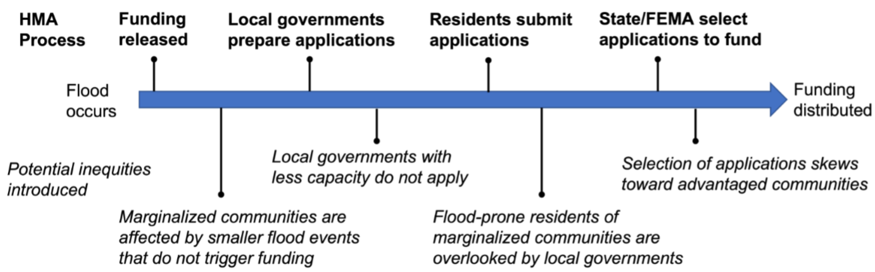

Figure 1: Inequitable outcomes across the FEMA Hazard Mitigation Assistance (HMA) application pipeline.

and administers the nation's largest risk mitigation programs, though other federal entities including the US Army Corps of Engineers, the National Oceanic and Atmospheric Administration, and the Environmental Protection Agency also play important roles.

Concerns about equitable access to FEMA assistance have been documented in the immediate aftermath of disasters for decades. Following presidential disaster declarations, FEMA's Individuals and Households Program (IHP) provides up to approximately $44,000 per household for temporary housing, property repair, unemployment assistance, and other immediate needs. Racial and income disparities in IHP access have been observed across a wide range of events, including the 1994 Northridge earthquake (Bolin and Stanford, 1998), Hurricane Katrina (**?**), Hurricane Dolly in South Texas (Rivera et al., 2022), and the 2017 Thomas Fire (Méndez et al., 2020). More broadly, studies find that social vulnerability is inversely associated with access to federal disaster recovery funds even after accounting for the extent of damage (Emrich et al., 2022; Drakes et al., 2021; Wilson et al., 2021; Bento and Elliott, 2022). Across disaster types and geographies, the pattern is consistent: federal assistance systematically underserves the communities most exposed to hazard (Raker, 2023).

Similar concerns have emerged around FEMA's risk mitigation programs, which fund long-term adaptation rather than immediate recovery. Typical Hazard Mitigation Assistance (HMA) projects include home elevations, property buyouts, and community-scale infrastructure; since 2020, the Building Resilient Infrastructure and Communities (BRIC) program has expanded community-wide project funding. Among property-scale programs, buyouts have received the most scrutiny. In a buyout, homeowners sell flood-prone properties to local governments, which demolish structures and restore the land to open space. Initiated in 1985 and now totaling over 43,000 completed transactions nationally, the buyout program has shifted from its rural origins into an increasingly urban policy (Mach et al., 2019). Its geographic distribution is deeply uneven: states with the highest cumulative flood damages (Florida, Louisiana, Mississippi) rank among the lowest for buyout deployment, while activity concentrates in wealthier, denser counties with greater administrative capacity (Mach et al., 2019). Within those counties, bought-out properties are concentrated in lower-income and more racially diverse neighborhoods, a pattern Mach et al. (2019) describe as a "buyout paradox" in which government capacity governs access at the county scale while within-county selection follows a different logic (Elliott et al., 2020; Loughran and Elliott, 2022). Home elevation funding exhibits a different and more troubling pattern, flowing disproportionately toward high-income and predominantly white communities (**?**, Frank 2022).

A structural driver of these disparities is the federal cost-benefit analysis (CBA) requirement. Because CBA estimates avoided losses as a function of property replacement value, projects in lower-income neighborhoods systematically generate lower benefit-to-cost ratios, making them less competitive for funding even when flood exposure is severe (Tate et al., 2021; Junod et al., 2021; Miller, 2023). The result is an equity trap in which the households most exposed and most vulnerable are simultaneously least likely to clear the threshold that determines who gets funded.

1.2. Procedural challenges to accessing funding

While these structural barriers shape outcomes at a broad scale, the mechanisms that produce inequity operate at the level of specific program rules and local implementation practices. The HMA application process is lengthy and multi-layered. After a presidential disaster declaration releases funding, state and local governments identify eligible properties, assist homeowners in compiling required documentation (including cost-benefit analyses, insurance records, and title documentation), and submit project proposals. Local governments ("sub-applicants") forward proposals to the state ("applicant"), which screens projects before submitting to FEMA for final review. The full process from disaster declaration to project closeout averages nearly six years (Mach et al., 2019). The federal government covers 75% of project costs; North Carolina covers the remaining 25% non-federal match, reducing a significant barrier that cash-strapped local governments face in other states.

Inequities can arise at every stage. At the county level, administrative capacity is the primary bottleneck: governments with dedicated staff, prior grant experience, and stronger institutional networks are far more effective at developing and submitting competitive proposals (Mach et al., 2019; Junod et al., 2021). At the community level, outreach practices vary widely; some local governments proactively contact eligible residents and guide them through the application process, while others do little or nothing, effectively limiting access to only those homeowners already aware of the program. At the household level, procedural requirements such as documentation, inspections, and legal title verification impose compliance burdens that fall disproportionately on lower-income households in under-resourced communities, mirroring documented barriers in IHP access (Rivera et al., 2022). Finally, CBA requirements further filter the pool: applications from lower-value properties, even if submitted, face systematic disadvantages (Miller, 2023).

Our analysis uses novel address-level data on the application and funding process to examine how each of these stages shapes the distribution of who ultimately benefits. By tracking properties from flood event to funding outcome, we can assess not just who receives mitigation assistance, but where along the pipeline the losses occur, and for whom.

2. Results

While most flooded properties are eligible and in communities that have applied for funding, very few properties submit applications (Fig. 2). Among all flooded properties (Fig. 2, left panel), we find that around 2/3 are eligible for federal funding (67.5%). Of these eligible properties, 89.9% are in communities where local government assisted with the submission of at least one application. Only a small fraction (3.8%) of eligible properties in applying communities apply themselves, equal to 2.3% of all flooded properties. Of those that apply, around half are funded (49.4%).

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